Roll No.: ………………….

Rayat-Bahra Institute of Engineering and Nano Technology

MST- I

Computer Networks

Semester: 5th CSE (A&B)
M. Marks: 24
Date:11/09/2024

Subject: CN(BTCS 504-18)
Time Allowed: 90 Mins
Session: Morning

Note: Section A is compulsory. Attempt any two questions from section B and One question from section C. Draw diagram where it is required

Section-A (2 marks each)

1. What is data communication? Explain its Flow networks.
2 What is Coaxial, Twisted and Fiber optic Cables?
3. Differentiate ARP and RARP.
4. What is Redundency,Error Correction and Error Detection?

Section-B (4 marks each)

5. What is Protocol? Explain any three protocol.
6. What is Selective Repeat ARQ Protocol?
7. Generate the CRC Code for data word

   M(X)=10111011 and Divisor=1001.

Section-C (8 marks each)

8. Explain OSI reference model with a neat diagram and also explain its layer functionality in detail.
9. Discuss in detail about different Network Topologies.

Wishing You All the Best!!

Roll No.: ………………

Rayat-Bahra Institute of Engineering and Nano Technology,Hoshiarpur

MST- I

SOFTWARE ENGINEERING

Branch/Semester: CSE/IT 5TH **Subject Code: (BTCS-503-18)(BTIT-504-18)**

M.Marks:24 **Time Allowed: 90 minutes**

Date: 10/09/24 **Session-Evening**

Note: Section A is compulsory. Attempt any two questions from section B and One question from section C. Draw diagram where it is required.

Section-A (2 marks each)

1. What is Software Program & Software Product?
2. Define Function Point Metric.
3. What is Software Engineering and its importance?
4. Explain the term Project Manager.

Section-B (4 marks each)

5. What are Project Planning and Control activities?
6. Differentiate between functional and non-functional requirements.
7. Explain LOC in Project Size Estimation.

Section-C (8 marks each)

8. What is the need of software life cycle models in Software Engineering? Explain Prototype Model of software development in detail.
9. Explain in detail the COCOMO Model of cost estimation.

Wishing You All the Best!!

Roll No.: ....

Rayat-Bahra Institute of Engineering and Nano Technology, Hoshiarpur

MST- I

DATABASE MANAGEMENT SYSTEM

Branch/Semester: CSE/IT 5TH&ME 7TH Subject Code:BTCS-(501-18)(BTIT-502-18)

M.Marks:24

Time Allowed: 90 minutes

Date: 10/09/24

Session-Morning

Note: Section A is compulsory Attempt any two questions from section B and One question from section C. Draw diagram where it is required.

Section-A (2 marks each)

1. State Augmentation rule with an example.
2. Explain DML commands with Syntax?
3. Draw the symbol of Derived and Multivalued attribute with example.
4. How will you differentiate between 3NF and BCNF?

Section-B (4 marks each)

5. Draw any ER Diagram which demonstrates the following:
   a. Entity
   b. Key Attribute
   c. Relationship
   d. Weak Entity
6. Differentiate relational alzebra and relational calculus.
7. State 2NF and convert the given table in 2NF

E.ID	COURSE	COURSE FEES
1	C1	1000
2	C2	1500
1	C4	2000
4	C3	1000

Section-C (8 marks each)

8. How will you explain outer join with Example.
9. In following relational model ,
   a. Find all the employees, working in same department by using Tuple calculus method.
   b. By using projection & selection operation find the employees whose salary >=20000

E.ID	NAME	SALARY	DEPT. NAME
1	ARASHDEEP	10000	SALES
2	RAJAT	20000	FINANCE
3	GURSHARAN	25000	SALES
4	RAHUL	28000	SALES

Roll No.: ....314..

Rayat-Bahra Institute of Engineering and Nanotechnology, Hoshiarpur

SPECIAL MST- I

Enterprise Resource Planning

Semester: CSE 5th
Subject: ERP (BTES 501-18)
M. Marks: 24
Time Allowed: 90 Mins
Date: 9 Sep 2024
Session: Evening

Note: Section A is compulsory. Attempt any two questions from section B and One question from section C. Draw diagram where it is required.

Section-A (2 marks each)

1. What is SCM & BRP?
2. Write down the Benefits of ERP.
3. Explain the term Hidden Cost concept with proper explanation.
4. What is the importance of Project Monitoring?

Section-B (4 marks each)

5. Define Data Mining & Data Warehouse? Write down the benefits of both of it.
6. Define OLAP. What are the 12 rules of OLAP?
7. Differentiate between ERP system and multiple software applications? Explain with the help of suitable example using suitable assumption if any.

Section-C (8 marks each)

8. Explain different phase of ERP Implementation life cycle. Explain with its example.

9. Explain the benefits of the ERP Package achieved at different levels of business modules.

Wishing You All the Best!!

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Rayat-Bahra Institute of Engineering and Nano Technology, Hoshiarpur

MST- I

Programming in Python

Semester: CSE 5th   Subject: Programming in Python (BTCS510-18)

M. Marks: 24   Time Allowed: 90 Mins

Date: 11 September 2024(Evening)

Note: Section A is compulsory. Attempt any two questions from section B and One question from section C. Draw diagram where it is required.

Section-A (2 marks each)

1. Write a short note on python language.
2. Define a Python module.
3. How to create a User-Defined function in python?
4. Write the code for "Convert the string to both uppercase and lowercase".
   text = "Learn Python"

Section-B (4 marks each)

5. List out the differences between a list, tuple, and dictionary in Python?
6. Explain the Operators that are supported by python.
7. Describe any two in-built functions and two in-built modules in Python.

Section-C (8 marks each)

8. Describe the process of exception detection and handling in detail with the help of an example. What role does assertion play in exception handling?
9. Explain in detail about Python Files, open() function and operations that can be performed on files with examples.

Wishing You All the Best!!

Rayat-Bahra Institute of Engineering and Nano Technology, Hoshiarpur

MST- I

Formal Language & Automata Theory

Semester: CSE/ IT 5th　　Subject: FLAT (BTCS 502-18/BTIT 501-18)

M. Marks: 24　　Time Allowed: 90 Mins

Date: 9 SEPT, 2024　　Session: Morning

Note: Section A is compulsory. Attempt any two questions from section B and One question from section C. Draw diagram where it is required.

Section-A (2 marks each)

1. State Arden's Theorem.
2. Define Derivation tree with its Types.
3. What are different applications of automata theory?
4. State pumping lemma for regular languages.

Section-B (4 marks each)

5. Write Regular Expressions for following statements:
   (i) All strings over {0, 1} with the substring '0101'.
   (ii) All strings over {1, a, b} beginning with the 11 and ending with ab.
   (iii) Set of all strings over {a, b} with 3 consecutive b.
   (iv) All strings over {a, b} having at least one b.
6. Describe DFA and NDFA in detail. Also give differences between them.
7. Explain in detail the Chomsky classification of languages.

Section-C (8 marks each)

8. Construct the minimum state automaton equivalent to the transition table:

State	a	b
→ q_0	q_1	q_2
q_1	q_4	q_3
q_2	q_4	q_3
q_3	q_5	q_6
q_4	q_7	q_6
q_5	q_3	q_6
q_6	q_6	q_6
q_7	q_4	q_6

9. Construct a DFA with reduced states equivalent to the regular expression:

$$(0+1)^* (00+11) (0+1)^*$$

Wishing You All the Best!!